

Proof H is Linearizable:

If multiple objects: Given the composability of linearizability, it suffices to show that for each object x , $H|x$ is linearizable. Proof each with steps:

Represent history H formally as: ... Let G denote the following complete history, and S denote the following sequential history.

Note that: G is obtained from H by discarding the pending invocation ... / appending the matching response ... for the pending invocation ...

We then have: $G|A = S|A = \dots$, $G|B = S|B = \dots$, ... and thus we know: G and S are equivalent. (1)

With the following S_p and S_q : ... Since for object p , S_p is in the sequential spec for p , and similarly for object q , S_q is in the sequential spec for q , we can show that:

S is a legal, sequential history of memory location objects. (2)

From the definition of $\rightarrow$ we have: $\rightarrow_G = \dots$, $\rightarrow_S = \dots$ and thus $\rightarrow_G \subseteq \rightarrow_S$. (3)

From (1), (2), (3) we can conclude that H is linearizable.